Zhang Zheng

The Hong Kong Polytechnic University
Hong Kong, China
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Education

The Hong Kong Polytechnic University (QS 54) — *2025 – present*
Ph.D. Student in Computing
Advisor: Prof. Jing Qin

Imperial College London (QS 2) — *2021 – 2022*
M.Sc. in Biomedical Engineering — High Merit
Advisor: Prof. Mengxing Tang

University of Nottingham (QS 97) — *2017 – 2021*
B.Sc. (Hons) in Computer Science with Artificial Intelligence — First Class Honours

Publications

Conference Publications

2. **FlowPET: Physics-Informed Symplectic Flow Matching for Low-Count PET Reconstruction**.
   Zhang Zheng, H. Tang, Y. Hu, Z. Hu, Jing Qin.
   *International Conference on Machine Learning (**ICML**), 2026.* **[CCF-A]**
1. **FourierPET: Deep Fourier-based Unrolled Network for Low-count PET Reconstruction**.
   Zhang Zheng, H. Tang, Y. Hu, Z. Hu, Jing Qin.
   *AAAI Conference on Artificial Intelligence (**AAAI**), 2026.* **[CCF-A, Oral Presentation, top 1.2%]**

Journal Publications

1. **Prompt-Agent-Driven Integration of Foundation Model Priors for Low-Count PET Reconstruction**.
   X. Xie, W. Zhao, M. Nan, **Zhang Zheng**, Y. Wu, H. Zheng, D. Liang, M. Wang, Z. Hu.
   *IEEE Transactions on Medical Imaging (**TMI**), 2025.* **[JCR Q1, IF=9.8)]**

Awards & Honours

- **Head's Scholarship**, University of Nottingham — *2019 – 2020*
  Awarded to the top 10% of students in the department.
- **Distinguished Final Year Project Award**, University of Nottingham — *2020 – 2021*
  Project: "Radar Human Activity Recognition Using Micro-Doppler Signature"

Research Experience

SIAT, Chinese Academy of Sciences — *2023 – 2025*
Research Assistant — Advisor: Prof. Zhanli Hu
Physics-informed generative modeling for low-count PET image reconstruction.

ULIS, Imperial College London — *2021 – 2022*
M.Sc. Research Project — Advisor: Prof. Mengxing Tang
Neural super-resolution for ultrasound localization microscopy (ULM).

University of Nottingham — *2020 – 2021*
B.Sc. Final Year Project (Distinguished) — Advisor: Prof. Jianfeng Ren
Micro-Doppler radar sensing for human gait recognition.

Research Interests

Medical AI, Medical Imaging, Brain-Computer Interfaces, Generative Artificial Intelligence for Healthcare.